



Advanced Mathematics

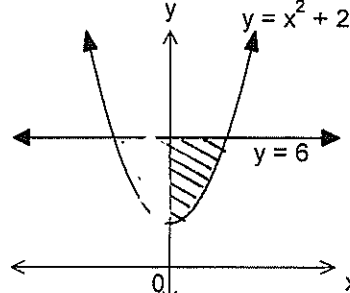
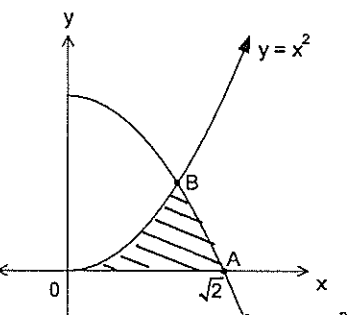
December 2010

Time: 50 minutes + 5 minutes reading time

DIRECTIONS

- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.
- Write your teacher's name and your name on the cover sheet provided
- At the end of the exam, staple your answers in order behind the cover sheet provided, and your questions on the back
- Approved Maths aids and calculators may be used

1.	Find a) $\int (3x - 7)^{15} dx$ b) $\int x\sqrt{x} dx$ c) $\int_0^5 \frac{7x^4 - 1}{x^2} dx$	2 2 3								
2.	Evaluate $\sum_{k=2}^5 (4k + 1)$	2								
3.	Archie invests \$1500 at 6.24% per year compound interest, compounding quarterly. Calculate the value of the investment after 5 years.	2								
4.	The sum of the first n terms of a certain arithmetic series is given by $S_n = \frac{n(3n+1)}{2}$ Find T_{17} .	2								
5.	Find the number which when added to each of 2, 6 and 13 will give the first 3 terms of a geometric progression.	3								
6.	The tenth term of an arithmetic sequence is 29, and the fifteenth term is 44. a) Find the value of the common difference and the value of T_1 . b) Find the sum of the first 75 terms.	3 2								
7.	For the function $f(x)$, $\int_1^6 f(x) dx = 7$ Hence, calculate $\int_1^6 (f(x) + 3) dx$	2								
8.	The following table lists the values of a function $y = f(x)$ for 3 values of x <table border="1"><tr><td>x</td><td>2.1</td><td>2.3</td><td>2.5</td></tr><tr><td>y</td><td>3</td><td>3.8</td><td>4.2</td></tr></table> a) Use Simpson's rule and these values to estimate $\int_{2.1}^{2.5} f(x) dx$ to 2 decimal places b) By using the trapezoidal rule with three function values, find the approximate volume of the solid generated when the area enclosed between the curve $y = f(x)$, the x -axis and the lines $x = 2.1$ and $x = 2.5$ is rotated about the x -axis. Give your answer in terms of π .	x	2.1	2.3	2.5	y	3	3.8	4.2	2 3
x	2.1	2.3	2.5							
y	3	3.8	4.2							
9.	a) Express $0.1\dot{4}$ as an infinite series b) Hence express $0.1\dot{4}$ as a fraction with no common factors.	1 2								

10.	To calculate the area of the region bounded by the curve $y = x^2 - 2x$ and the x-axis between $x = 0$ and $x = 3$, Harry used $\int_0^3 x^2 - 2x \, dx$ a) Explain why Harry's method of calculating this area is incorrect. 1 b) Find the required area. 3
11.	 In the diagram, the shaded region is bounded by the parabola $y = x^2 + 2$, the y-axis and the line $y = 6$ Find the volume of the solid formed when the shaded region is rotated about the y-axis 3
12.	 The shaded region OAB is bounded by the parabolas $y = x^2$ and $y = 2 - x^2$ and the x-axis between $x = 0$ and $x = \sqrt{2}$ a) Find the coordinates of B 1 b) Calculate the area OAB (2 decimal places) 3
13.	a) Joe invests a fixed amount, P at the beginning of each year into a superannuation fund for 10 years. The compound interest rate is 4%pa, compounded annually. By establishing a series, find the amount P, invested if the value of the superannuation account at the end of the 10th year was \$89900, to the nearest dollar. 4 b) Joe then retires and intends to draw a set amount M from this fund each month. He finds a bank that offers him 4.8%pa on his lump sum compounding monthly and he expects to withdraw M for the next 15 years. i) Show that at the end of the 2nd month the amount remaining is 1 $A_2 = 89900(1.004)^2 - M(1.004 + 1)$ ii) Hence calculate M to the nearest dollar by first showing that 3 $M = \frac{359.6(1.004)^{180}}{1.004^{180} - 1}$
~ END OF EXAM ~	

$$1. a) \int (3x-7)^{15} dx = \frac{(3x-7)^{16}}{3 \times 16} + C$$

$$= \frac{(3x-7)^{16}}{48} + C$$

$$b) \int x^{\frac{3}{2}} dx = \frac{2}{5} x^{\frac{5}{2}} + C$$

$$c) \int_0^5 7x^2 - x^{-2} dx = \left[\frac{7x^3}{3} + \frac{1}{x} \right]_0^5$$

$$= \left(\frac{7 \times 5^3}{3} + \frac{1}{5} \right) - \left(\frac{7}{3} + 1 \right)$$

$$= 288.53$$

$$2. \sum_{k=2}^6 (4k+1) = 9 + 13 + 17 + 21$$

$$= 60$$

$$3. A_{20} = 1500 \left(1 + \frac{0.0624}{4} \right)^{20}$$

$$= \$2044.30$$

$$4. S_n = \frac{n(3n+1)}{2}$$

$$S_{17} = \frac{17(3 \times 17 + 1)}{2} \quad S_{16} = \frac{16(3 \times 16 + 1)}{2}$$

$$T_{17} = S_{17} - S_{16}$$

$$= 442 - 392$$

$$= 50.$$

$$5. 2+x, 6+x, 13+x \text{ --- C.P.}$$

$$\frac{6+x}{2+x} = \frac{13+x}{6+x} \quad (1)$$

$$(6+x)^2 = (13+x)(2+x)$$

$$36 + 12x + x^2 = 26 + 15x + x^2 \quad (1)$$

$$10 = 3x$$

$$x = \frac{10}{3} \quad (1)$$

$$6. T_{10} = a + 9d = 29 \quad (1)$$

sub into T_{10}

$$T_{15} = a + 14d = 44$$

$$a + 27 = 29$$

$$a = 2 \quad (1)$$

$$a) T_{15} - T_{10} = 5d = 15$$

$$d = 3 \quad (1)$$

$$\therefore T_1 = 2, d = 3.$$

$$7. \int_1^b f(x) dx = 7$$

$$\int_1^b (f(x) + 3) dx = 7 + [3x]_1^b \quad (1)$$

$$= 7 + (18 - 3) \quad (2)$$

$$= 22. \quad (1)$$

$$6b1$$

$$S_{75} = \frac{75}{2} (2 \times 2 + 74 \times 3) \quad (1)$$

$$= 8475 \quad (1)$$

$$8. a) \int_{2.1}^{2.5} f(x) dx = \frac{0.2}{3} \{ 3 + 4 \times 3.8 + 4.2 \} \quad (1)$$

$$= 1.4933 \dots$$

$$\approx 1.49 \text{ (2dp)} \quad (1)$$

y	3	3.8	4.2
y ²	9	14.44	17.64

$$V = \pi \int_{2.1}^{2.5} y^2 dx \quad (1)$$

$$= \pi \times \frac{0.2}{2} \times \{ 9 + 2 \times 14.44 + 17.64 \} \quad (1)$$

$$= 5.552 \pi \text{ v}^3 \quad (1)$$

$$1. a) 0.1\dot{4} = 0.1 + \frac{4}{100} + \frac{4}{1000} + \frac{4}{10000} + \dots$$

$$\text{or } 0.1 + 0.04 + 0.004 + 0.0004 + \dots$$

①

$$b) \lim S = \frac{a}{1-r}$$

$$= \frac{\frac{4}{100}}{1 - \frac{1}{10}}$$

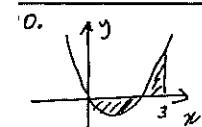
$$= \frac{4}{90} \quad \text{①}$$

$$\therefore 0.1\dot{4} = \frac{1}{10} + \frac{4}{90}$$

$$= \frac{13}{90} \quad \text{①}$$

③

some may use other method
(still allocate 2 marks)



a) Area under x axis is neg and cancels
out area above curve

①
something
logical

$$b) A = \left| \int_0^2 x^2 - 2x \, dx \right| + \int_2^3 x^2 - 2x \, dx$$

$$= \left| \left[\frac{x^3}{3} - x^2 \right]_0^2 \right| + \left[\frac{x^3}{3} - x^2 \right]_2^3$$

$$= \left| \frac{8}{3} - 4 - 0 \right| + \left(9 - 9 - \left(\frac{8}{3} - 4 \right) \right) \quad \text{①}$$

$$= \frac{4}{3} + \frac{4}{3}$$

$$= \frac{8}{3} \text{ u}^2 \quad \text{①}$$

④

$$11. V = \pi \int_2^6 y - 2 \, dy \quad \text{①}$$

$$= \pi \left[\frac{y^2}{2} - 2y \right]_2^6 \quad \text{①}$$

$$= \pi (18 - 12 - (2 - 4))$$

$$= 8\pi \text{ u}^3 \quad \text{①}$$

or dec answer

$$25.13 \dots$$

③

12.

$$a) y = x^2 \dots \text{①}$$

$$y = 2 - x^2 \dots \text{②}$$

①

$$x^2 = 2 - x^2$$

$$\therefore B = (1, 1)$$

$$x^2 = 1$$

$$x = \pm 1$$

$$b) A = \int_0^1 x^2 \, dx + \int_1^{\sqrt{2}} 2 - x^2 \, dx \quad \text{①}$$

$$= \left[\frac{x^3}{3} \right]_0^1 + \left[2x - \frac{x^3}{3} \right]_1^{\sqrt{2}} \quad \text{①}$$

$$= \frac{1}{3} - 0 + \left(2\sqrt{2} - \frac{2\sqrt{2}}{3} - \left(2 - \frac{1}{3} \right) \right)$$

$$= \frac{1}{3} + \frac{4\sqrt{2} - 5}{3}$$

$$= \frac{4\sqrt{2} - 4}{3} \quad \text{①}$$

$$= 0.55 \text{ u}^2$$

④

13.

$$a) A_1 = P(1.04)^{10} \quad A_2 = P(1.04)^9 \quad A_3 = P(1.04)^8 \dots$$

$$A_{10} = P(1.04)$$

$$\therefore \text{Total } A = P(1.04 + 1.04^2 + 1.04^3 + \dots + 1.04^{10}) \quad \text{①}$$

G.P

$$a = 1.04 \quad r = 1.04 \quad n = 10$$

$$\text{① } 89900 = P \times \frac{1.04(1.04^{10} - 1)}{1.04 - 1}$$

$$P = 7199.86 \dots$$

$$\approx \$7200 \quad \text{①}$$

$$b) r = 4.8\% \text{ pa} \quad n = 180$$

$$i) = 0.004$$

$$A_1 = 89900(1.004) - M \quad \text{①}$$

$$A_2 = A_1(1.004) - M \rightarrow 89900(1.004)^2 - (M \times 1.004) - M$$

$$= 89900(1.004)^2 - M(1.004 + 1)$$

$$\text{② } A_3 = 89900(1.004)^3 - M(1.004^2 + 1.004 + 1)$$

$$\therefore A_{180} = 89900(1.004)^{180} - M(1 + 1.004 + \dots)$$

$$\text{but } A_{180} = 0$$

$$0 = 89900(1.004)^{180} - M \left(\frac{1(1.004^{180} - 1)}{1.004 - 1} \right)$$

$$M = 89900(1.004)^{180} \times \frac{0.004}{1.004^{180} - 1}$$

$$= 359.6(1.004)^{180} / 1.004^{180} - 1 = \$701.59$$

⑧